司南导航

Product Specification / 产品规范

# K803 and K823 Adapter Board

# K803, K823 转接板

2020-12-29

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# REVISION HISTORY / 修订历史

| REVISION / 版本 | MODIFICATION / 更改 | DATE / 日期 |
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| 1.2 | Add electrical description. / 增加电气说明 | 2020-12-29 |
| 1.1 | Add dimensional description. / 增加尺寸说明 | 2020-12-14 |
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# I. INTRODUCTION / 简介

K803、K823 are high precision positioning multi-system OEM Modules based on a self-developed SoC, it has small size, multi-system and multi-frequency. They track constellations including GPS, BDS-2, BDS-3, GLONASS, Galileo, SBAS and QZSS, are mainly designed and used for UAV and handheld devices with size, weight, and power requirements.

In order to be convenient for using, this specification especially notes the corresponding adapter board.

K803、K823 模块是司南导航自主研发的多系统、多频点、小尺寸高精度定位、定向 OEM 模块，满足多系统多频点导航卫星系统模块需求，支持 GPS、BDS-2、BDS-3、GLONASS、Galileo 以及 SBAS 和 QZSS。适用于如无人机和手持设备等对模块尺寸、重量和功耗要求严格的领域。

为了更好地方便用户使用，本规范着重介绍对应规格转接板。

# II. SPECIFICATION OF K803&K823 ADAPTER BOARD / K803&K823 转接板技术规范

Following table presents the detailed specification of ComNav K803&K823 Adapter Board. Specific technical characteristics are listed.

下表中为司南 K803&K823 转接板的各项技术性能。

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*Table 1. K803&K823 Adapter Board Specification*

| K803&K823 ADAPTER BOARD SPECIFICATION/ K803&K823 转接板规范 | | | |
|---|---|---|---|
| **Signals**<br>**信号** | K803 | Positioning 定位 | GPS: L1, L1C, L2C, L2P, L5 |
| | | | BDS-2: B1I, B2I, B3I<br>BDS-3: B1I, B3I, B1C, B2a, B2b* |
| | | | GLONASS: G1, G2 |
| | | | Galileo: E1, E5a, E5b |
| | | | QZSS: L1, L2C, L5 |
| | | | SBAS: L1 |
| | | | IRNSS: L5* |
| | | | L-band* |
| | K823 | Positioning 定位 | GPS: L1, L2 |
| | | | BDS-2: B1I, B3I<br>BDS-3: B1I, B3I, B1C, B2b* |
| | | | GLONASS: G1, G2 |
| | | | Galileo: E1, E5b |
| | | | QZSS: L1, L2* |
| | | | SBAS: L1 |
| | | | L-band* |
| | | Orientation 定向 | GPS: L1, L2 |
| | | | BDS-2: B1I, B3I<br>BDS-3: B1I, B3I |
| | | | GLONASS: G1, G2 |
| | | | Galileo: E1, E5b |
| | | | QZSS: L1, L2* |
| | 带*项会随同版本进行调整 | | |

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| K803&K823 ADAPTER BOARD SPECIFICATION/ K803&K823 转接板规范 | | |
|---|---|---|
| **Time to First Fix**<br>**首次定位时间** | Cold Start<br>冷启动 | < 30s（Adding Acceleration Capture Module,增加捕获加速模块） |
| | Hot Start (with RTC)<br>热启动（使用 RTC） | < 15s（Typical,典型） |
| **Reacquisition**<br>**信号重捕** | 失锁重捕 | < 1s |
| **Measurement Precision**<br>**测量准确度** | Pseudorange Precision<br>伪距精度 | ≤ 10cm |
| | Carrier Phase Precision<br>载波相位精度 | ≤ 0.005c (c: carrier wave length of GNSS signal, in meter) (c: 载波波长，单位：米) |
| **Accuracy**<br>**精度** | Time Accuracy<br>授时精度 | 20ns |
| | SPP Accuracy<br>标准单点定位精度 | H ≤ 1.5m, V ≤ 3m (1σ, PDOP≤4) |
| | Static Differential Accuracy<br>(Supported by Compass Solution)<br>静态差分精度（Compass Solution 软件支持） | H: ±(2.5+1×$10^{-6}$×D) mm<br>V: ±(5.0+1×$10^{-6}$×D) mm<br>D 为基线长度(单位：km)<br>D - Baseline length (Unit: km) |
| | Speed accuracy<br>测速精度 | ≤ 0.02 m/s (PDOP ≤4) |
| **Inertial Navigation (Optional)**<br>**惯导（选配）** | While the GNSS antenna signal is losing lock for 3 seconds, the accuracy maintains at centimeter level.<br>While the GNSS antenna signal is losing lock for 10 seconds, the accuracy maintains at meter level.<br>GNSS 天线信号失锁 3s，精度保持 cm 级 | |

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| K803&K823 ADAPTER BOARD SPECIFICATION/ K803&K823 转接板规范 | | |
|---|---|---|
| | GNSS 天线信号失锁 10s，精度保持 m 级 | |
| **Anti-interference 抗干扰** | It can suppress the potential narrowband and single tone radio interference signals in the GNSS signal frequency band, and the interference to signal ratio can reach 50dB.<br>具备抑制 GNSS 信号频带内潜在的窄带和单音无线电干扰信号，干信比可达 50dB。 | |
| **Attitude Accuracy 测姿精度（K823）** | Azimuth Accuracy<br>方位角精度 | (0.2/R) °, R is baseline length in meter.<br>R 为基线长度，单位为米 |
| | Roll or Pitch Accuracy<br>横滚或俯仰角 | (0.4/R) °, R is baseline length in meter.<br>R 为基线长度，单位为米 |
| **RTK** | RTK Initialization time<br>RTK 初始化时间 | < 10s（baseline < 10km，基线长小于 10km） |
| | Initialization Reliability<br>初始化置信度 | > 99.9 % |
| | RTK Accuracy<br>RTK 精度 | H: ± (8 + $10^{-6}$ × D) mm<br>V: ± (15 + $10^{-6}$ × D) mm<br>D 为基线长度(单位：km)<br>D - Baseline length (Unit: km) |
| **Data Rates 数据速率** | Measurements & Position<br>测量&定位 | 20Hz（Optional，选配项） |
| | RTK: Positioning & Heading<br>RTK：定位定向<br>（定向仅 K823 支持）<br>（Only K823 support heading） | 20Hz（Optional，选配项） |
| **Environmental 环境要求** | Operating Temperature<br>工作温度 | -40℃ — +85℃ |
| | Storage Temperature<br>储存温度 | -55℃ — +95℃ |
| **Data Formats** | NMEA-0183 | GPGGA, GPGSV, GPGLL, GPGSA, GPGST, GPHDT, |

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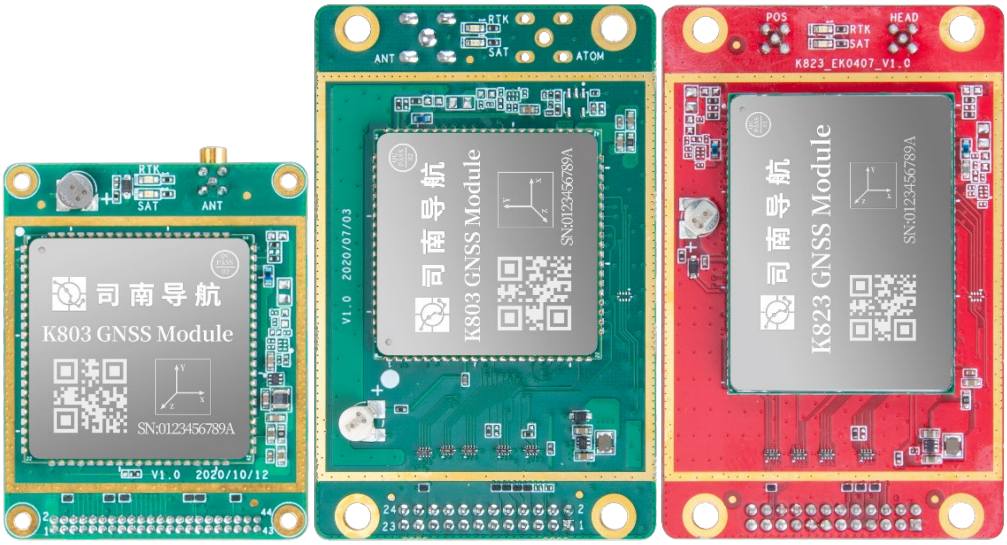
| K803&K823 ADAPTER BOARD SPECIFICATION/ K803&K823 转接板规范 | | |
|---|---|---|
| **输出数据格式** | | GPRMC, GPVTG, GPZDA etc. |
| | ComNav Binary（CNB）<br>司南二进制格式 | ComNav Self-Defined Binary<br>司南自定义二进制 |
| | CMR(GPS) | CMROBS, CMRREF |
| | RTCM2.X | RTCM1, RTCM3, RTCM9, RTCM1819, RTCM31, RTCM41, RTCM42 |
| | RTCM3.X | 1004 ~ 1008, 1012, 1019, 1020, 1033, 1042, 1045/1046, 1230, 4078<br>MSM3~MSM7: 1073~1077, 1083~1087, 1123~1127, 1093~1097 |
| **Antenna Interface 天线接口** | LNA Power<br>天线供电电压 | External 外部供电:<br>+3.3V ~ +5V ± 5%VDC @ 0-100mA |
| | LNA Gain<br>天线增益要求 | 20 ~ 40dB （Suggested 建议） |
| **Electrical 电气特性** | Voltage<br>供电电压 | +3.3V~5.5V DC |
| | Power consumption<br>功耗 | K803:1.0W(Anti-interference off，抗干扰关闭)<br>K823:1.6W(Anti-interference off，抗干扰关闭)<br>Anti-interference on may consume 0.2W more.<br>开启抗干扰约增加 0.2W。 |

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# III. DIMENSION / 尺寸

In this section, product photo, three-side views and the dimension of K803 / K823 are provided for customers' further hardware design and installation.

本节提供了 K803_EK0405、K803_EK0407、K823_EK0407、K803_EK0610 的实物图。




K803_EK0405　　K803_EK0407　　K823_EK0407

K803_EK0610

*Figure 1. Product Photo*

图 1. 转接板 1:1 实物图

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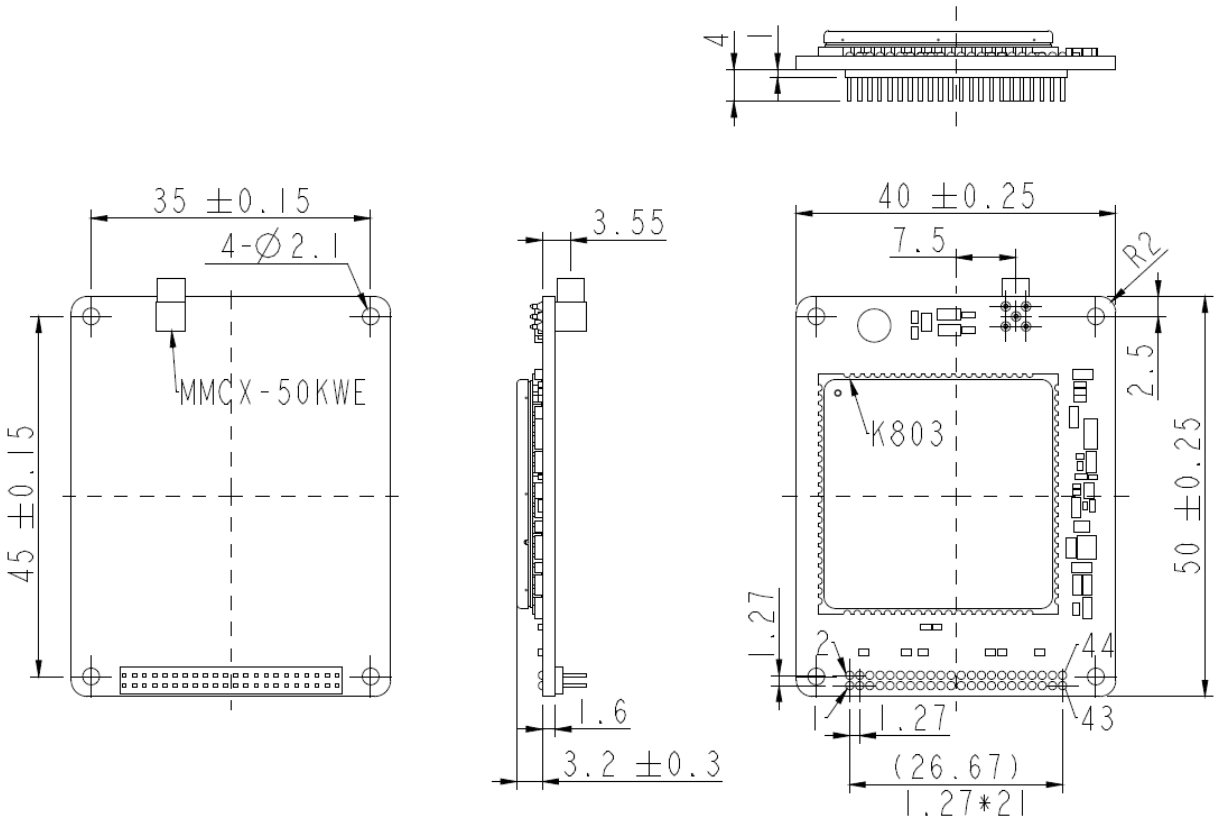



*Figure 2. Dimensions of K803_EK0405*

图 2. K803_EK0405 尺寸图

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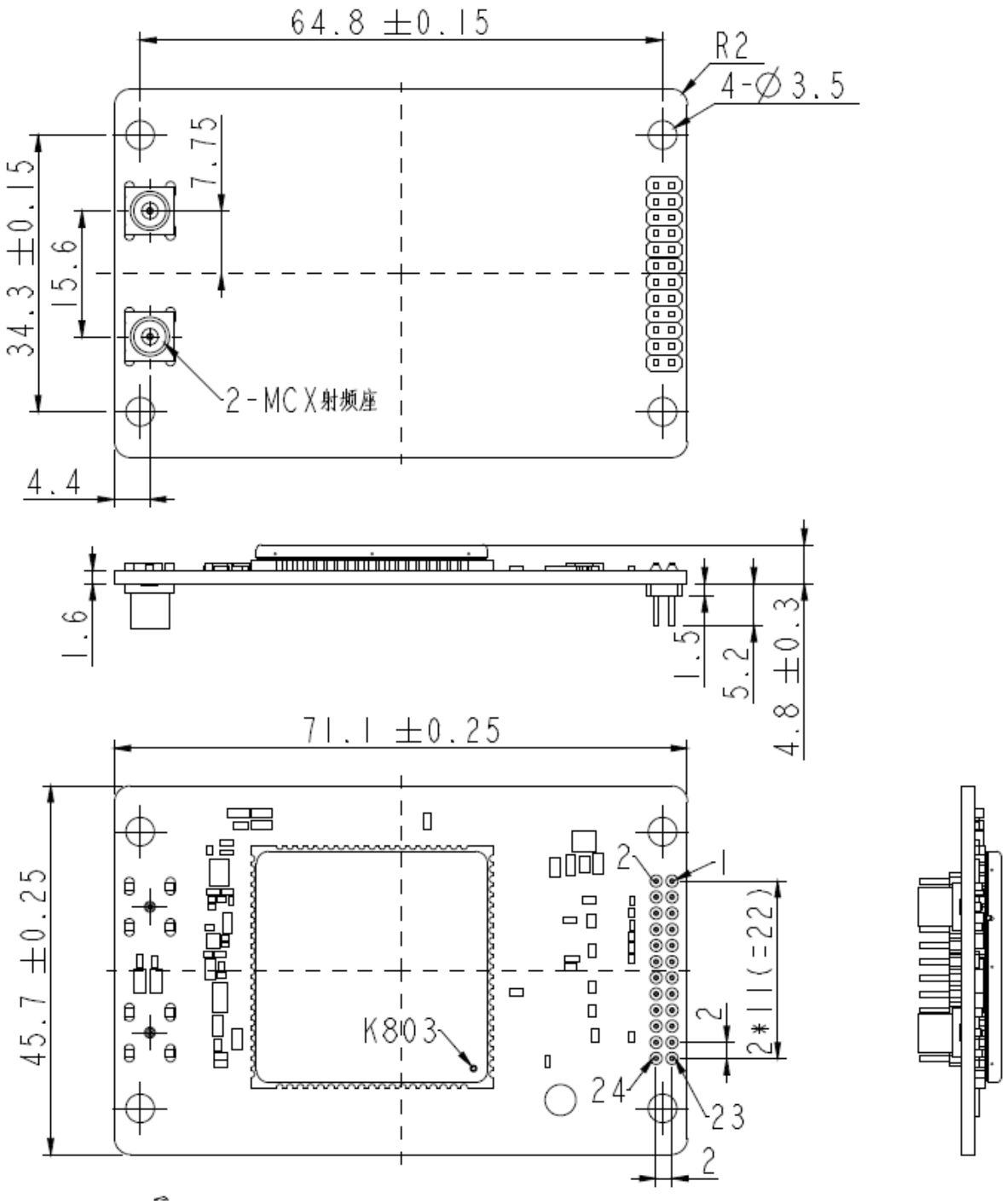



*Figure 3. Dimensions of K803_EK0407*

图 3. K803_EK0407 尺寸图

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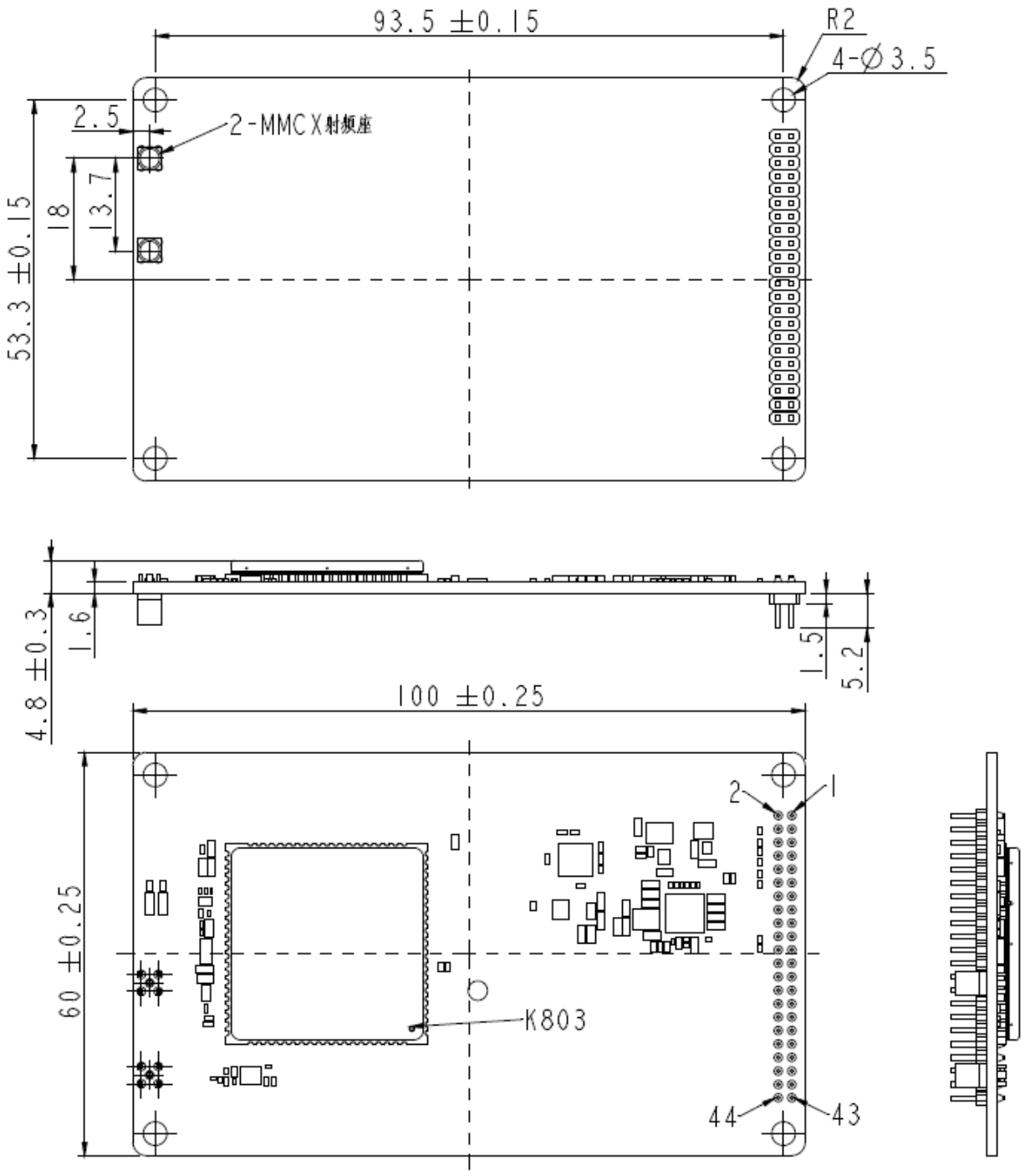



*Figure 4. Dimensions of K803_EK0610*

图 4. K803_EK0610 尺寸图

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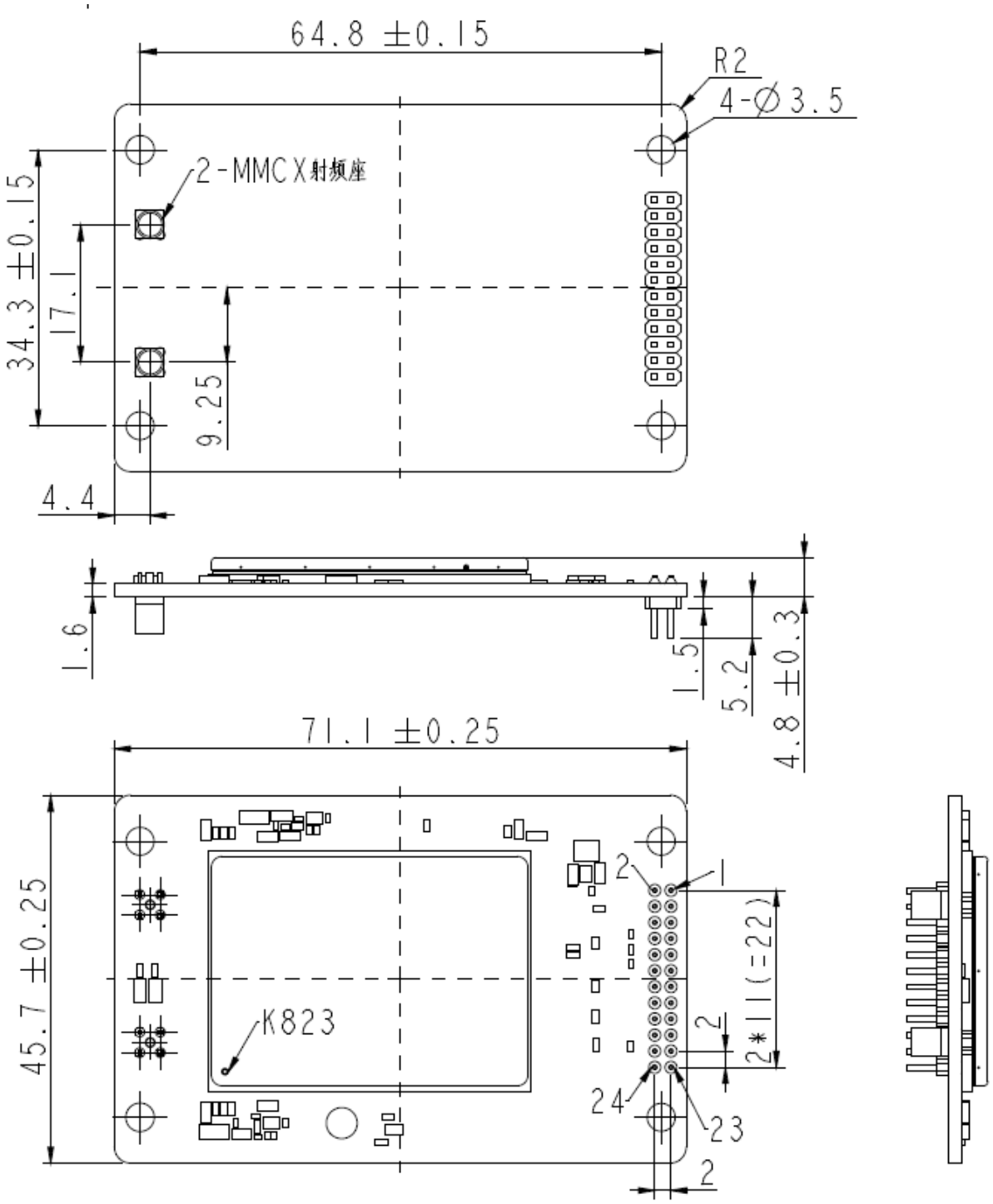



*Figure 5. Dimensions of K823_EK0407*

图 5. K823_EK0407 尺寸图

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# IV. PIN ARRANGEMENT AND DEFINITION OF K803_EK0405 BOARD/ K803_EK0405 引脚标识和定义

K803_EK0405 has one 44-Pin connector (22 Pin, 1.27mm Dual Row vertical T/H HDR), is compatible with K705.

K803_EK0405 包括 44 针连接头（针脚间距 1.27mm，双排），与 K705 兼容。

*Table 2. Pin Definition of K803_EK0405 44-Pin Connector*

| PIN | SIGNAL | TYPE | DESCRIPTION | |
|---|---|---|---|---|
| 1 | VIN | PWR | Power supply for board | 板卡供电 |
| 2 | VIN | PWR | Power supply for board | 板卡供电 |
| 3 | GND | GND | Ground reference | 参考地 |
| 4 | Reserved | - | Reserved | 保留 |
| 5 | GND | GND | Ground reference | 参考地 |
| 6 | SYS_RST | I | System reset | 系统复位信号 |
| 7 | COM1_TX | O | UART1 TX | COM1 串口输出 |
| 8 | COM1_RX | I | UART1 RX | COM1 串口输入 |
| 9 | COM2_TX | O | UART2 TX | COM2 串口输出 |
| 10 | COM2_RX | I | UART2RX | COM2 串口输入 |
| 11 | COM3_TX | O | UART3 TX | COM3 串口输出 |
| 12 | COM3_RX | I | UART3 RX | COM3 串口输入 |
| 13 | GND | GND | Ground reference | 参考地 |
| 14 | LNA_PWR | PWR | Power supply for external LNA | 天线供电 |
| 15 | PPS | O | Pulse output synchronized to GNSS time | 同步卫星时间脉冲 |
| 16 | EVENT2 | I | Event2 mark | 外部事件 2 输入 |
| 17 | COM4_TX | O | UART4 TX | COM4 串口输出 |
| 18 | COM4_RX/PV | I | UART4 RX/Position valid indicator | COM4 串口输入/定位有效指示<br>默认：COM4 串口输入 |

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| PIN | SIGNAL | TYPE | DESCRIPTION | |
|---|---|---|---|---|
| | | | Default: UART4 RX | |
| 19 | LED_RTK | O | RTK data indicator | RTK 数据指示灯 |
| 20 | LED_SAT | O | Tracked satellite number indicator | 跟踪卫星数量指示灯 |
| 21 | SPI_CS | O | SPI CS | SPI 片选信号 |
| 22 | SPI_CLK | O | SPI Clock | SPI 时钟信号 |
| 23 | SPI_MOSI | I/O | SPI_MOSI | SPI 主输出从输入 |
| 24 | SPI_MISO | I/O | SPI_MISO | SPI 主输入从输出 |
| 25 | GND | GND | Ground reference | 参考地 |
| 26 | Reserved | - | Reserved | 保留 |
| 27 | Reserved | - | Reserved | 保留 |
| 28 | Reserved | - | Reserved | 保留 |
| 29 | Reserved | - | Reserved | 保留 |
| 30 | Reserved | - | Reserved | 保留 |
| 31 | GND | GND | Ground reference | 参考地 |
| 32 | Reserved | - | Reserved | 保留 |
| 33 | Reserved | - | Reserved | 保留 |
| 34 | Reserved | - | Reserved | 保留 |
| 35 | Reserved | - | Reserved | 保留 |
| 36 | Reserved | - | Reserved | 保留 |
| 37 | EVENT1 | I | Event1 mark | 外部事件 1 输入 |
| 38 | Reserved | - | Reserved | 保留 |
| 39 | Reserved | - | Reserved | 保留 |
| 40 | Reserved | - | Reserved | 保留 |
| 41 | Reserved | - | Reserved | 保留 |
| 42 | Reserved | - | Reserved | 保留 |

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| PIN | SIGNAL | TYPE | DESCRIPTION | |
|---|---|---|---|---|
| 43 | Reserved | - | Reserved | 保留 |
| 44 | Reserved | - | Reserved | 保留 |

***REMARKS / 说明:***

*1.* Electrical characteristics /电气特性

COM1_TX, COM1_RX, COM2_TX, COM2_RX, COM3_TX, COM3_RX, COM4_TX, COM4_RX, SPI_CLK, SPI_CS, SPI_MOSI and SPI_MISO are LVCMOS 3.3V. SYS_RST, LED_RTK, LED_SAT, PPS, EVENT, EVENT2 are LVTTL 3.3V. All these signals are compatible with LVCMOS/LVTTL 3.3V.

COM1_TX, COM1_RX, COM2_TX, COM2_RX, COM3_TX, COM3_RX, COM4_TX, COM4_RX, SPI_CLK, SPI_CS, SPI_MOSI以及SPI_MISO为LVCMOS 3.3V电气标准。SYS_RST, LED_RTK, LED_SAT, PPS, EVENT, EVENT2为LVTTL 3.3V电平，所有这些信号均兼容LVCMOS/LVTTL 3.3V。

**LVCMOS 3.3V电气标准**

| Symbols 符号 | Description 描述 | Min 最小 | Max 最大 |
|---|---|---|---|
| **VIH** | Input high voltage<br>输入高电压 | 2V | 3.6V |
| **$V_{IL}$** | Input low voltage<br>输入低电压 | -0.3V | 0.8V |
| **$V_{OH}$** | High-level output voltage<br>高电平输出电压 | 2.9V | -- |
| **$V_{OL}$** | Low-level output voltage<br>低电平输出电压 | -- | 0.4V |
| **$I_{OH}$** | Sourcing current<br>拉电流 | 8mA | |
| **$I_{OL}$** | Sinking current<br>灌电流 | 8mA | |

| Symbols/符号 | Description/描述 | Min/最小 | Max/最大 |
|---|---|---|---|
| VIH | Input high voltage<br>输入高电压 | 2.0V | —— |
| **$V_{IL}$** | Input low voltage<br>输入低电压 | -0.3V | 0.8V |
| **$V_{OH}$** | High-level output voltage<br>高电平输出电压 | 2.4V | ---- |
| **$V_{OL}$** | Low-level output voltage<br>低电平输出电压 | --- | 0.4V |
| **$I_{OH}$** | Sourcing current | 8mA | |

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| Symbols/符号 | Description/描述 | Min/最小 | Max/最大 |
|---|---|---|---|
| | 拉电流 | | |
| $I_{OL}$ | Sinking current<br>灌电流 | 8mA | |

2. Signals whose absolute maximum rating is -0.3V ~ 3.6V are as follows:

   所能承受电压的最大值范围是-0.3V ~ 3.6V的信号如下：COM1_TX, COM1_RX, COM2_TX, COM2_RX, COM3_TX, COM3_RX, COM4_TX, COM4_RX, SYS_RST, SPI_CLK, SPI_CS, SPI_MOSI and SPI_MISO.

3. VCC

   Main power supply, voltage range is 3.3V DC ~ 5.5V DC.

   The requirement for voltage ripple and spike is less than 100mV.

   主供电电源，电压范围：3.3V ~ 5.5V（直流）。电压纹波和尖峰脉冲要求小于100mV。

4. LED_RTK / LED_SAT

   LED_RTK flashing indicates that data from base station is received. LED_SAT is used to indicate the satellite number, whose flashing times at one time represents the captured satellite number. Both LED_RTK and LED_SAT are all high active. External LED driver is needed for normal use.

   LED_RTK闪烁指示接收到基准站的数据，LED_SAT指示卫星数量，一次连续闪烁的次数表示当前搜到卫星的数量。LED_RTK与LED_SAT均为高电平驱动LED，需要外加LED驱动。

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# V. PIN ARRANGEMENT AND DEFINITION OF K803_EK0407 BOARD/ K803_EK0407 引脚标识和定义

K803_EK0407 has one 24-Pin connector (12 Pin, 2mm Dual Row vertical T/H HDR), is compatible with K706/K707.

K803_EK0407 包括 24 针连接头（针脚间距 2mm，双排），与 K706/K707 兼容。

*Table 3. Pin Definition of K803_EK0407 24-Pin Connector*

| PIN | SIGNAL | TYPE | DESCRIPTION | |
|---|---|---|---|---|
| 1 | SPI_CLK | O | SPI Clock | SPI 时钟信号 |
| 2 | SPI_CS | O | SPI CS | SPI 片选信号 |
| 3 | LNA_PWR | PWR | Power supply for external LNA | 天线供电 |
| 4 | VIN | PWR | Power supply for board | 板卡供电 |
| 5 | Reserved | - | Reserved | 保留 |
| 6 | COM3_RX | I | UART3 RX | COM3 串口输入 |
| 7 | RST_SYS | I | System reset | 系统复位信号 |
| 8 | VARF | O | 10MHz square wave output | 10MHz 方波输出 |
| 9 | EVENT2 | I | Event2 mark | 外部事件 2 输入 |
| 10 | LED_RTK | O | RTK data indicator | RTK 数据指示灯 |
| 11 | COM3_TX/EVENT1 | I/O | UART3 TX/ Event1 mark<br>Default: UART3 TX | COM3 串口输出/外部事件 1 输入<br>默认：COM3 串口输出 |
| 12 | GND | GND | Ground Reference | 参考地 |
| 13 | COM1_TX | O | UART1 TX | COM1 串口输出 |
| 14 | COM1_RX | I | UART1 RX | COM1 串口输入 |
| 15 | GND | GND | Ground Reference | 参考地 |
| 16 | COM2_TX | O | UART2 TX | COM2 串口输出 |
| 17 | COM2_RX | I | UART2 RX | COM2 串口输入 |
| 18 | GND | GND | Ground Reference | 参考地 |

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| PIN | SIGNAL | TYPE | DESCRIPTION | |
|---|---|---|---|---|
| 19 | GPIO | I/O | GPIO | 通用 IO |
| 20 | GND | GND | Ground Reference | 参考地 |
| 21 | PPS | O | Pulse output synchronized to GNSS time | 同步卫星时间脉冲 |
| 22 | LED_SAT | O | Tracked satellite number indicator | 跟踪卫星数量指示灯 |
| 23 | SPI_MISO | I/O | SPI_MISO | SPI 主输入从输出 |
| 24 | SPI_MOSI | I/O | SPI_MOSI | SPI 主输出从输入 |

***REMARKS/说明:***

*5.* Electronic characteristic/电气特性

LED_RTK, LED_SAT, COM1_TX, COM1_RX, COM2_TX, COM2_RX, COM3_TX, COM3_RX, RST_SYS, SPI_CLK, SPI_CS, SPI_MOSI, SPI_MISO, are LVCMOS 3.3V.

LED_RTK, LED_SAT, COM1_TX, COM1_RX, COM2_TX, COM2_RX, COM3_TX, COM3_RX, RST_SYS, SPI_CLK, SPI_CS, SPI_MOSI, SPI_MISO, 为LVCMOS 3.3V电气标准。

**LVCMOS 3.3V电气标准**

| Symbols 符号 | Description 描述 | Min 最小 | Max 最大 |
|---|---|---|---|
| VIH | Input high voltage<br>输入高电压 | 2V | 3.6V |
| $V_{IL}$ | Input low voltage<br>输入低电压 | -0.3V | 0.8V |
| $V_{OH}$ | High-level output voltage<br>高电平输出电压 | 2.9V | -- |
| $V_{OL}$ | Low-level output voltage<br>低电平输出电压 | -- | 0.4V |
| $I_{OH}$ | Sourcing current<br>拉电流 | 8mA | |
| $I_{OL}$ | Sinking current<br>灌电流 | 8mA | |

*6.* PPS, EVENT2 and VARF are LVTTL 3.3V and compatible with LVCMOS/LVTTL 3.3.

PPS, EVENT2和VARF为LVTTL 3.3 V电平，所有这些信号兼容LVCMOS / LVTTL 3.3V。

| Symbols/符号 | Description/描述 | Min/最小 | Max/最大 |
|---|---|---|---|
| VIH | Input high voltage | 2.0V | —— |

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| Symbols/符号 | Description/描述 | Min/最小 | Max/最大 |
|---|---|---|---|
| | 输入高电压 | | |
| $V_{IL}$ | Input low voltage<br>输入低电压 | -0.3V | 0.8V |
| $V_{OH}$ | High-level output voltage<br>高电平输出电压 | 2.4V | ---- |
| $V_{OL}$ | Low-level output voltage<br>低电平输出电压 | --- | 0.4V |
| $I_{OH}$ | Sourcing current<br>拉电流 | 8mA | |
| $I_{OL}$ | Sinking current<br>灌电流 | 8mA | |

7. Absolute maximum rating is -0.3V~3.6V of following signals:/所能承受电压的最大值范围是-0.3V~3.6V的信号如下：

   LED_RTK, LED_SAT, COM1_RX, COM1_TX, COM2_RX, COM2_TX, COM3_RX, COM3_TX, EVENT2, PPS, VARF, RST_SYS, SPI_CLK, SPI_CS, PI_MOSI, SPI_MISO.

8. VIN

   Main power supply, voltage range is 3.3VDC~5.5V DC.

   Voltage ripple and spike requirement: <100mV

   主供电电源（输入），电压范围：3.3V~5.5V（直流）。电压纹波和尖峰脉冲需求：<100mV。

9. RET_SYS

   Low active, it can be used to reset the whole OEM board.

   低电平有效，可用于复位整个OEM板。

10. LED_RTK/LED_SAT

    LED_SAT is used to indicate the satellite number. LED_RTK indicates that RTK is undergoing. Both LED_RTK and LED_SAT are high active. External LED driver is needed for normal use.

    LED_RTK闪烁指示接收到基准站的数据，LED_SAT指示卫星数量，一次连续闪烁的次数表示当前搜到卫星的数量。LED_RTK与LED_SAT均为高电平驱动LED，需要外加LED驱动。

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# VI. PIN ARRANGEMENT AND DEFINITION OF K823_EK0407 BOARD/ K823_EK0407 引脚标识和定义

K823_EK0407 has one 24-Pin connector (12 Pin, 2mm Dual Row vertical T/H HDR), is compatible with K726.

K823_EK0407 板卡包括 24 针连接头（针脚间距 2mm，双排），兼容 K726。

*Table 4. Pin Definition of K823_EK0407 24-Pin Connector*

| PIN | SIGNAL | TYPE | DESCRIPTION | |
|---|---|---|---|---|
| 1 | SPI_CLK | O | SPI Clock | SPI 时钟信号 |
| 2 | SPI_CS | O | SPI CS | SPI 片选信号 |
| 3 | LNA_PWR | PWR | Power supply for external LNA | 天线供电 |
| 4 | VIN | PWR | Power supply for board | 板卡供电 |
| 5 | Reserved | - | Reserved | 保留 |
| 6 | COM3_RX | I | UART3 RX | COM3 串口输入 |
| 7 | RST_SYS | I | System reset | 系统复位信号 |
| 8 | Reserved | - | Reserved | 保留 |
| 9 | EVENT2 | I | Event2 mark | 外部事件 2 输入 |
| 10 | LED_RTK | O | RTK data indicator | RTK 数据指示灯 |
| 11 | COM3_TX/EVENT1 | I/O | UART3 TX/ Event1 mark<br>Default: UART3 TX | COM3 串口输出/外部事件 1 输入<br>默认：COM3 串口输出 |
| 12 | GND | GND | Ground Reference | 参考地 |
| 13 | COM1_TX | O | UART1 TX | COM1 串口输出 |
| 14 | COM1_RX | I | UART1 RX | COM1 串口输入 |
| 15 | GND | GND | Ground Reference | 参考地 |
| 16 | COM2_TX | O | UART2 TX | COM2 串口输出 |
| 17 | COM2_RX | I | UART2 RX | COM2 串口输入 |
| 18 | GND | GND | Ground Reference | 参考地 |

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| PIN | SIGNAL | TYPE | DESCRIPTION | |
|---|---|---|---|---|
| 19 | GPIO | I/O | GPIO | 通用 IO |
| 20 | GND | GND | Ground Reference | 参考地 |
| 21 | PPS | O | Pulse output synchronized to GNSS time | 同步卫星时间脉冲 |
| 22 | LED_SAT | O | Tracked satellite number indicator | 跟踪卫星数量指示灯 |
| 23 | SPI_MISO | I/O | SPI_MISO | SPI 主输入从输出 |
| 24 | SPI_MOSI | I/O | SPI_MOSI | SPI 主输出从输入 |

***REMARKS/说明:***

1. Electronic characteristic/电气特性

   LED_RTK, LED_SAT, COM1_TX, COM1_RX, COM2_TX, COM2_RX, COM3_TX, COM3_RX, RST_SYS, SPI_CLK, SPI_CS, SPI_MOSI, SPI_MISO, are LVCMOS 3.3V.

   LED_RTK, LED_SAT, COM1_TX, COM1_RX, COM2_TX, COM2_RX, COM3_TX, COM3_RX, RST_SYS, SPI_CLK, SPI_CS, SPI_MOSI, SPI_MISO, 为LVCMOS 3.3V电气标准。

   **LVCMOS 3.3V电气标准**

| Symbols 符号 | Description 描述 | Min 最小 | Max 最大 |
|---|---|---|---|
| VIH | Input high voltage<br>输入高电压 | 2V | 3.6V |
| $V_{IL}$ | Input low voltage<br>输入低电压 | -0.3V | 0.8V |
| $V_{OH}$ | High-level output voltage<br>高电平输出电压 | 2.9V | -- |
| $V_{OL}$ | Low-level output voltage<br>低电平输出电压 | -- | 0.4V |
| $I_{OH}$ | Sourcing current<br>拉电流 | 8mA | |
| $I_{OL}$ | Sinking current<br>灌电流 | 8mA | |

2. PPS, EVENT2 and VARF are LVTTL 3.3V All these signals are compatible with LVCMOS/LVTTL 3.3.

   PPS, EVENT2和VARF为LVTTL 3.3 V电平，所有这些信号兼容LVCMOS / LVTTL 3.3V。

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| Symbols/符号 | Description/描述 | Min/最小 | Max/最大 |
|---|---|---|---|
| VIH | Input high voltage<br>输入高电压 | 2.0V | —— |
| $V_{IL}$ | Input low voltage<br>输入低电压 | -0.3V | 0.8V |
| $V_{OH}$ | High-level output voltage<br>高电平输出电压 | 2.4V | ---- |
| $V_{OL}$ | Low-level output voltage<br>低电平输出电压 | --- | 0.4V |
| $I_{OH}$ | Sourcing current<br>拉电流 | 8mA | |
| $I_{OL}$ | Sinking current<br>灌电流 | 8mA | |

3. Absolute maximum rating is -0.3V~3.6V of following signals:/所能承受电压的最大值范围是-0.3V~3.6V的信号如下：

   LED_RTK, LED_SAT, COM1_RX, COM1_TX, COM2_RX, COM2_TX, COM3_RX, COM3_TX, EVENT2, PPS, VARF, RST_SYS, SPI_CLK, SPI_CS, PI_MOSI, SPI_MISO.

4. VIN

   Main power supply, voltage range is 3.3VDC~5.5V DC.

   Voltage ripple and spike requirement: <100mV

   主供电电源（输入），电压范围：3.3V~5.5V（直流）。电压纹波和尖峰脉冲需求：<100mV。

5. RST_SYS

   Low active, it can be used to reset the whole OEM board, which is 3.0V compatible.

   低电平有效，可用于复位整个OEM板。

6. LED_RTK/LED_SAT

   LED_SAT is used to indicate the satellite number. LED_RTK indicates that RTK is undergoing. Both LED_RTK and LED_SAT are all high active. External LED driver is needed for normal use.

   LED_RTK闪烁指示接收到基准站的数据，LED_SAT指示卫星数量，一次连续闪烁的次数表示当前搜到卫星的数量。LED_RTK与LED_SAT均为高电平驱动LED，需要外加LED驱动。

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# VII. PIN ARRANGEMENT AND DEFINITION OF K803_EK0610 BOARD/ K803_EK0610 引脚标识和定义

K803_EK0610 has one 44-Pin connector (22 Pin, 2mm Dual Row vertical T/H HDR), is compatible with K708.

K803_EK0610 板卡包括 44 针连接头（针脚间距 2mm，双排），兼容 K708。

*Table 5. Pin Definition of K803_EK0610 44-Pin Connector*

| PIN | SIGNAL | TYPE | DESCRIPTION | |
|---|---|---|---|---|
| 1 | GND | PWR | Ground Reference | 参考地 |
| 2 | LED_RTK | O | RTK data LED indicator | RTK 数据指示灯 |
| 3 | Reserved | - | Reserved | 保留 |
| 4 | PPS | O | Pulse output synchronized to GNSS Time | 同步卫星时间脉冲 |
| 5 | VIN | PWR | Power supply for board | 系统供电电源 |
| 6 | VIN | PWR | Power supply for board | 系统供电电源 |
| 7 | COM3_RX | I | UART3 RX | COM3 串口输入 |
| 8 | EVENT1 | I | Event1 mark | 外部事件输入 1 |
| 9 | Reserved | - | Reserved | 保留 |
| 10 | LED_SAT | O | Tracked satellite number indicator | 跟踪卫星数量指示灯 |
| 11 | VARF | O | 10MHz square wave output | 10MHz 方波输出 |
| 12 | RET_SYS | IO | System reset | 系统复位 |
| 13 | GPIO | IO | GPIO | 通用 IO |
| 14 | COM2_RX | I | UART2 RX | COM2 串口输入 |
| 15 | COM4_RX | I | UART4 RX | COM4 串口输入 |
| 16 | COM2_TX | O | UART2 TX | COM2 串口输出 |
| 17 | COM4_TX | O | UART4 TX | COM4 串口输出 |
| 18 | COM1_RX | I | UART1 RX | COM1 串口输入 |

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| PIN | SIGNAL | TYPE | DESCRIPTION | |
|---|---|---|---|---|
| 19 | COM3_TX | O | UART3 TX | COM3 串口输出 |
| 20 | COM1_TX | O | UART1 TX | COM1 串口输出 |
| 21 | Reserved | - | Reserved | 保留 |
| 22 | Reserved | - | Reserved | 保留 |
| 23 | GND | PWR | Ground reference | 参考地 |
| 24 | GND | PWR | Ground reference | 参考地 |
| 25 | SPI_CLK | O | SPI clock signal | SPI 总线时钟信号 |
| 26 | SPI_MISO | I | SPI MISO signal | SPI 主输入信号 |
| 27 | Reserved | - | Reserved | 保留 |
| 28 | Reserved | - | Reserved | 保留 |
| 29 | Reserved | - | Reserved | 保留 |
| 30 | Reserved | - | Reserved | 保留 |
| 31 | Reserved | - | Reserved | 保留 |
| 32 | Reserved | - | Reserved | 保留 |
| 33 | Reserved | - | Reserved | 保留 |
| 34 | Reserved | - | Reserved | 保留 |
| 35 | GND | PWR | Ground reference | 参考地 |
| 36 | Reserved | - | Reserved | 保留 |
| 37 | Reserved | - | Reserved | 保留 |
| 38 | Reserved | - | Reserved | 保留 |
| 39 | Reserved | - | Reserved | 保留 |
| 40 | Reserved | - | Reserved | 保留 |
| 41 | Reserved | - | Reserved | 保留 |
| 42 | GND | PWR | Ground reference | 参考地 |
| 43 | SPI_CS0 | O | SPI CS signal | SPI 总线片选信号 |
| 44 | SPI_MOSI | O | SPI M0SI signal | SPI 主输出信号 |

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***REMARKS / 说明:***

1. Electronic characteristic/电气特性

   LED_RTK, LED_SAT, COM2_TX, COM2_RX, COM3_TX, COM3_RX, RST_SYS, SPI_CLK, SPI_CS, SPI_MOSI and SPI_MISO are LVCMOS 3.3V. COM4_TX and COM4_RX are LVCMOS 1.8V

   LED_RTK, LED_SAT, COM2_TX, COM2_RX, COM3_TX, COM3_RX, RST_SYS, SPI_CLK, SPI_CS, SPI_MOSI, SPI_MISO为LVCMOS 3.3V电气标准。COM4_TX, COM4_RX为LVCMOS 1.8V电气标准。

   **LVCMOS 3.3V电气标准**

| Symbols 符号 | Description 描述 | Min 最小 | Max 最大 |
|---|---|---|---|
| VIH | Input high voltage<br>输入高电压 | 2V | 3.6V |
| $V_{IL}$ | Input low voltage<br>输入低电压 | -0.3V | 0.8V |
| $V_{OH}$ | High-level output voltage<br>高电平输出电压 | 2.9V | -- |
| $V_{OL}$ | Low-level output voltage<br>低电平输出电压 | -- | 0.4V |
| $I_{OH}$ | Sourcing current<br>拉电流 | 8mA | |
| $I_{OL}$ | Sinking current<br>灌电流 | 8mA | |

   **LVCMOS 1.8V电气标准**

| Symbols 符号 | Description 描述 | Min 最小 | Max 最大 |
|---|---|---|---|
| VIH | Input high voltage<br>输入高电平 | 1.3V | 2.1V |
| $V_{IL}$ | Input low voltage<br>输入低电平 | -0.3V | 0.4V |
| $V_{OH}$ | High-level output voltage<br>高电平输出电压 | 1.4V | -- |
| $V_{OL}$ | Low-level output voltage<br>低电平输出电压 | -- | 0.4V |
| $I_{OH}$ | Sourcing current<br>拉电流 | 8mA | |
| $I_{OL}$ | Sinking current<br>灌电流 | 8mA | |

2. PPS, EVENT and VARF are LVTTL 3.3V and compatible with LVCMOS/LVTTL 3.3.

   PPS、EVENT和VARF为LVTTL 3.3 V电平，所有这些信号兼容LVCMOS / LVTTL 3.3V。

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| Symbols/符号 | Description/描述 | Min/最小 | Max/最大 |
|---|---|---|---|
| VIH | Input high voltage<br>输入高电压 | 2.0V | —— |
| $V_{IL}$ | Input low voltage<br>输入低电压 | -0.3V | 0.8V |
| $V_{OH}$ | High-level output voltage<br>高电平输出电压 | 2.4V | ---- |
| $V_{OL}$ | Low-level output voltage<br>低电平输出电压 | --- | 0.4V |
| $I_{OH}$ | Sourcing current<br>拉电流 | 8mA | |
| $I_{OL}$ | Sinking current<br>灌电流 | 8mA | |

3. Absolute maximum rating is -0.3V~3.6V of following signals:/所能承受电压的最大值范围是-0.3V~3.6V的信号如下：

   LED_RTK, LED_SAT, COM2_RX, COM2_TX, COM3_RX, COM3_TX, COM4_RX, COM4_TX, EVENT, PPS, VARF, RST_SYS, SPI_CLK, SPI_CS, PI_MOSI, SPI_MISO.

4. COM1_TX and COM1_RX are RS232 compatible and can communicate with computer directly. COM4_TX and COM4_RX are LVCMOS 1.8V.

   COM1_TX和COM1_RX 为RS232电平，可直接与电脑通讯。COM4_TX 和COM4_RX是LVCMOS 1.8V。

5. VIN

   Main power supply, voltage range is 3.3V DC~5.5V DC.

   Voltage ripple and spike requirement: <100mV

   主供电电源（输入），电压范围：3.3V~5.5V（直流）。电压纹波和尖峰脉冲需求：<100mV。

6. RST_SYS

   Low active, it can be used to reset the whole OEM board.

   低电平有效，可用于复位整个OEM板。

7. LED_RTK/LED_SAT

   LED_SAT is used to indicate the satellite number. LED_RTK indicates that RTK is undergoing. Both LED_RTK and LED_SAT are high active. External LED driver is needed for normal use.

   LED_RTK闪烁指示接收到基准站的数据，LED_SAT指示卫星数量，一次连续闪烁的次数表示当前搜到卫星的数量。LED_RTK与LED_SAT均为高电平驱动LED，需要外加LED驱动。

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